

Supervised Learning Algorithms

PART A • CONCEPTUAL UNDERSTANDING



1. What are Supervised Learning Algorithms?

Supervised Learning is a machine learning technique where the model learns from labeled data.

It learns the relationship between input and output to make predictions on new data.

Example: Predicting house prices from previous house data.



Figure 1: Labeled data trains the algorithm, which produces a model that predicts new outputs.

2. Difference between Regression and Classification Algorithms

Regression is used to predict continuous numerical values, such as salary or house price.

Classification is used to predict categories or classes, such as Pass/Fail or Spam/Not Spam.

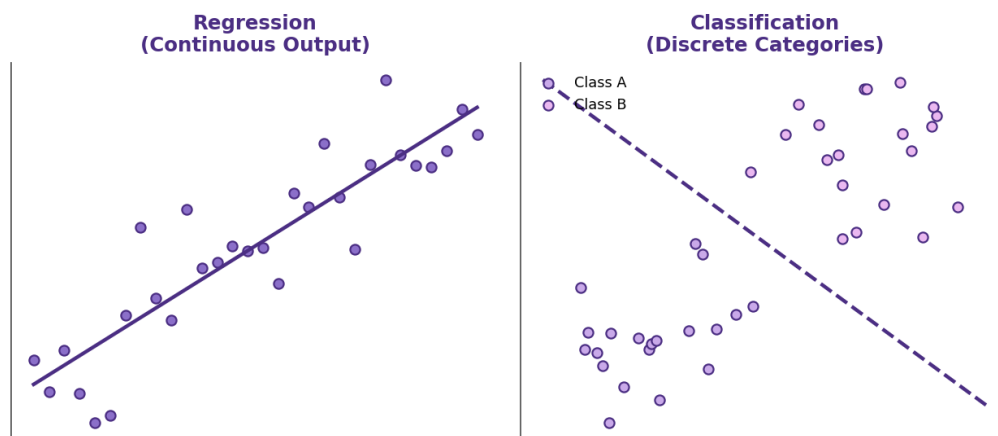


Figure 2: Regression predicts continuous values; Classification predicts discrete categories.

3. Simple Linear Regression

Simple Linear Regression is used to predict a dependent variable using one independent variable.

It assumes a linear relationship between X and Y.

$$Y = \beta_0 + \beta_1 X + \varepsilon$$

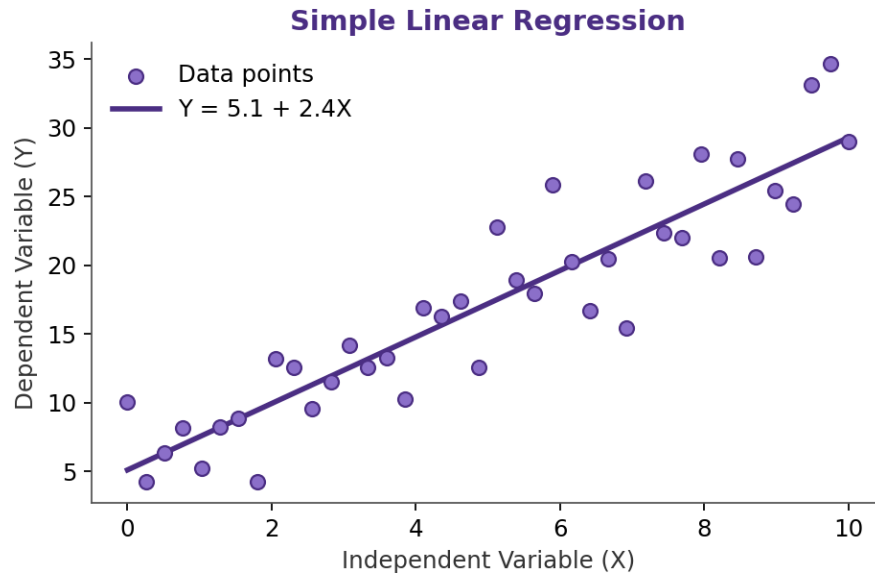


Figure 3: A best-fit line describing the linear relationship between X and Y .

4. Assumptions of Linear Regression

- **Linearity:** X and Y should have a linear relationship.
- **Independence:** Observations should be independent.
- **Homoscedasticity:** Error variance should remain constant.
- **Normality:** Errors should be approximately normally distributed.
- **No significant outliers:** Extreme values should not strongly affect the model.

5. Bias–Variance Trade-Off

Bias–Variance Trade-Off means finding the right balance between Bias and Variance for better model performance.

High Bias causes Underfitting, while high Variance causes Overfitting.

The goal is to build a model that generalizes well to new data.

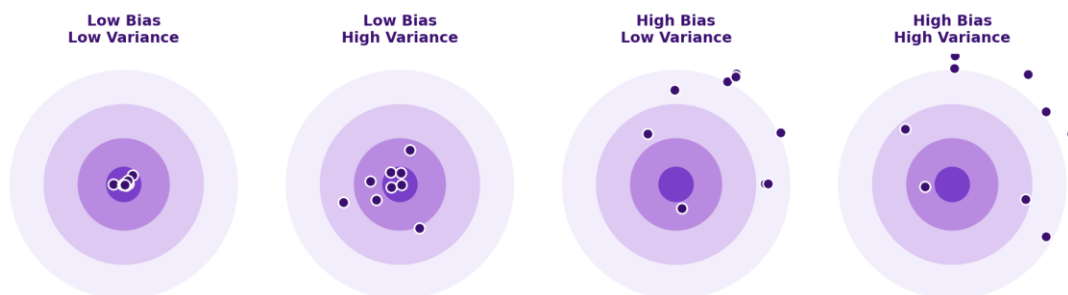


Figure 4: The dartboard analogy — bias measures how far off-target predictions are; variance measures how scattered they are.

6. Overfitting and Underfitting

Overfitting

Overfitting occurs when a model learns training data too closely, including noise, and performs poorly on new data.

Example: A very complex polynomial model fitting every training point.

Underfitting

Underfitting occurs when a model is too simple to learn the actual pattern.

Example: Using a straight line for data having a strong curved relationship.

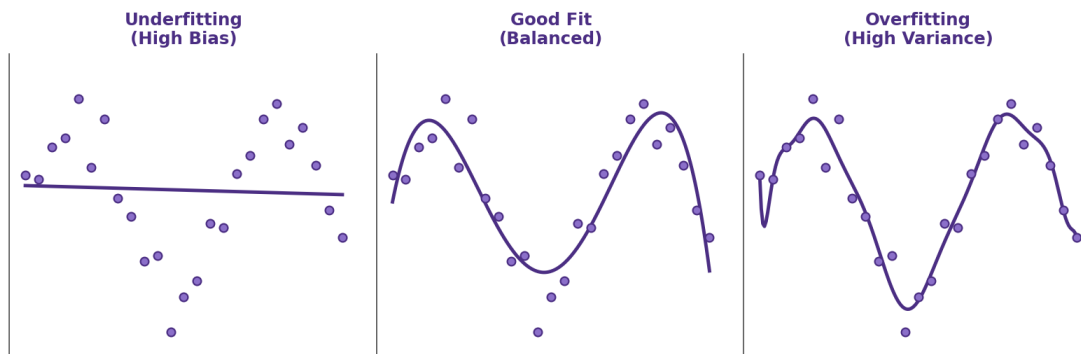


Figure 5: Underfitting (high bias) vs. a balanced fit vs. Overfitting (high variance).